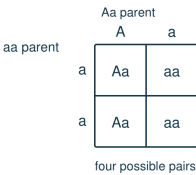


I am learning to calculate genetic probability with a Punnett square and evaluate its limits.

Words to use

| Word           | Meaning                                                   | Example                                              | My language |
|----------------|-----------------------------------------------------------|------------------------------------------------------|-------------|
| allele         | a version of a gene                                       | A and a represent two alleles in our invented model. |             |
| genotype       | the pair of alleles shown by letters                      | Aa is one possible genotype in this model.           |             |
| dominant       | an allele shown by one copy in this simple model          | A is dominant in our invented flower model.          |             |
| recessive      | an allele shown when two copies are present in this model | aa shows the recessive version in our model.         |             |
| Punnett square | a grid that shows possible allele pairs                   | The Punnett square for Aa × aa has four cells.       |             |

Model



Invented A/a flower model: Aa × aa gives two Aa and two aa possibilities. Each square is a possibility, not a family count. Simplified.

Place A/a atop the grid for one parent; place a/a at its side for the other. Pair one allele from each edge: Aa, aa, Aa, aa. Two of four boxes are aa: a 1 in 2 chance per offspring, not an exact family count.

How I know

- I can pair one allele from each parent.
- I can find the aa chance in four squares.
- I can evaluate why probability does not guarantee an exact count.



You can plan in any language. Speak before you write: The chance is ...  
because ...; this does not promise ...